SUMAIYA BINTE HARUN

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EDUCATION

Institute of Business Administration (IBA), University of Dhaka — Dhaka, Bangladesh
Master of Business Administration (MBA); CGPA: 3.20/4.00 — 2023 – Jan 2027 (expected)

- Coursework completed in business analytics, research methods, operations, finance and strategic management; internship report in progress.

Khulna University of Engineering & Technology (KUET) — Khulna, Bangladesh
Bachelor of Science in Civil Engineering; CGPA: 3.02/4.00 (final 60 credits: 3.23/4.00) — 2016 – 2020

- Thesis: Analysis of Surface Water Quality Adjacent to the Waste Disposal Site in Khulna. Supervisor: Dr. Md. Rafizul Islam.

RESEARCH INTERESTS

Behaviour and performance of reinforced and prestressed concrete bridges • Bridge assessment, residual capacity, repair and rehabilitation • Bridge substructures, foundations and soil–structure interaction • Infrastructure resilience and lifecycle decision-making

PROFESSIONAL EXPERIENCE

Design Planning & Management (DPM) Consultants Ltd. — Dhaka, Bangladesh
Junior Structural Engineer (Bridge Design) — June 2021 – Present

- Perform structural analysis and detailed design of RCC and PSC I-girder bridges and flyovers (80+ structures to date, including culverts) for Government of Bangladesh agencies including the Local Government Engineering Department (LGED), the Roads and Highways Department (RHD) and city corporations.
- Develop and analyse bridge superstructure and substructure models in MIDAS Civil under dead, live, seismic, wind and braking loads to AASHTO, the Bangladesh National Building Code (BNBC) and LGED/RHD standards.
- Design all bridge components (girders, deck, piers, pier caps, abutments, wing walls, pile caps, piles), modelling soil–structure interaction with SPT-based soil springs and checking structural and geotechnical pile capacity.
- Select bridge types and span arrangements considering loading, hydrology, subsoil conditions, constructability and site constraints, informed by site visits to project locations.
- Present and defend designs before LGED and RHD design units, revising them in response to client review.
- Prepare design calculations and technical reports, supervise general arrangement and reinforcement drawings, and coordinate with geotechnical, hydrology and drafting teams.

SELECTED PROJECT EXPERIENCE

Chittagong Division Important Upazila & Union Road Widening and Strengthening Project (CDWSP)
Client: Local Government Engineering Department (LGED), Government of Bangladesh · 2025 – Present (ongoing)

- Planning, structural analysis, detailed design and drawing supervision of RCC and PSC I-girder bridges within a programme of approximately 180 bridges, mostly replacing old, narrow bridges to suit widened roads.

Consultancy Services for Comprehensive Feasibility Study to Develop Hathazari Junction by Means of Construction of Bypass (At-Grade) or Grade Separated Intersection for Chattogram Road Division
Client: Roads and Highways Department (RHD), Government of Bangladesh · Late 2024 – Early 2025

- Contributed to the structural design of two PSC I-girder flyovers, 1,520 m (38 × 40 m spans) and 1,840 m (46 × 40 m spans), working on assigned portions: superstructure modelling and pier and foundation design, considering seismic effects, construction staging and serviceability.
- Took part in design reviews comparing girder spacing, pier configurations and construction sequencing for cost and constructability.

Feasibility Study and Detailed Design of Narrow, Old, and Dilapidated Bridges (RCC, PSC & PSB) on Various Roads under Barisal Region
Client: Roads and Highways Department (RHD), Government of Bangladesh · 2024

- Feasibility assessment and detailed structural design of RCC and PSC I-girder replacement bridges within a programme of 138 structurally deficient and functionally obsolete bridges; optimised layouts on narrow alignments to limit land acquisition and traffic disruption.

Feasibility Study and Detail Structural Design for Improvement of Road and Bridges in Chittagong City Corporation
Client: Chittagong City Corporation (CCC), local government authority · 2023

- Contributed to the feasibility study and structural design of 17 bridges (PSC I-girder, RCC girder and slab; 10–40 m spans) and 22 culverts under urban constraints such as utilities, drainage and restricted working space.

Feasibility Study for the Sunamganj–Polash–Barikkatila–Takerghat–Mohishkhola–Durgapur Border Road Project
Associated authority: 34 Engineer Construction Brigade, Bangladesh Army · 2023

- Approximately 110 km border road. Contributed to route selection and preliminary technical assessment considering topography, hydrology and transport conditions, including hill runoff, heavy stone- and sand-carrying traffic and long-

term maintenance; assessed suitable bridge types and spans, and an embankment-cum-road (causeway) arrangement for sections with heavy water flow.

Mymensingh Region Rural Infrastructure Development Project (MRRIDP)
Client: Local Government Engineering Department (LGED), Government of Bangladesh · June 2021 – 2022

- Structural analysis and design within a programme of 135 PSC I-girder, RCC girder and slab bridges (10–50 m spans), adapting standardised designs to site-specific soil and waterway conditions.

TECHNICAL SKILLS

Analysis & design: MIDAS Civil, STAAD.Pro, ETABS **Drafting & BIM:** AutoCAD, Revit
Data: Python (used in MBA analytics project), Power BI, Advanced Excel **Codes:** AASHTO, BNBC, LGED and RHD specifications

PUBLICATIONS (CONFERENCE PROCEEDINGS)

- **Harun, S. B.**, & Rafizul, I. M. (2020). Analysis of surface water quality adjacent to the waste disposal site in Khulna. *Proceedings of the 5th International Conference on Civil Engineering for Sustainable Development (ICCESD 2020)*, KUET, Bangladesh. [click here]
- Mehzabeen, F., Hassan, K. M., Dey, S., & **Harun, S. B.** (2023). Development of trickling filter for wastewater treatment. *Proceedings of WasteSafe 2023*, Khulna, Bangladesh, p. 393. [click here]
- Azad, M. A., Adhikary, S. K., & **Harun, S. B.** (2023). Drought vulnerability assessment using fuzzy based approach: A case study on northwest region of Bangladesh. Presented at *International Perspective on Water Resources and the Environment (IPWE 2023)*, Dhaka, Bangladesh. [click here]

TEACHING & MENTORING

Instructor, Knowledge Session on RCC Bridge Design & Introduction to MIDAS Civil, KUET May 2023

- Taught 80+ final-year civil engineering students bridge planning, AASHTO loads and pile design, with a hands-on MIDAS Civil modelling demonstration.

PROFESSIONAL MEMBERSHIP & TRAINING

Member, Institution of Engineers, Bangladesh (IEB), No. M52607
Training: DPM technical sessions (CPD, 3 h each): Pile Static Load Test Interpretation (2026); USCS & Liquefaction (2026) • Bangladesh Public Procurement Rules (21 h) • Python Programming, DataCamp (28 h) • AutoCAD 2D & 3D (2 weeks, CADers, KUET) • Revit (2 weeks, BECM Association, Department of Building Engineering and Construction Management, KUET) • ETABS (2 weeks, Civil Association, KUET)

LEADERSHIP & SERVICE

KUET Debating Society — Assistant General Secretary 2019 – 2020

- Organised inter-university, divisional and national debate tournaments and about 10 campus programmes with 400+ participants.

Spectrum, Professional Skill Development Club of KUET — Chief Policy Analyst 2018 – 2019

- Led six events and trained participants in four workshops and seminars with 450+ participants.

Institution of Civil Engineers (ICE), KUET Chapter — Publication & Publicity Secretary 2018 – 2019

Viqarunnisa Noon Debating Club — Vice President 2018 – 2019

Volunteers for Bangladesh — Member Jan 2025 – Present

- Volunteer in community programmes for underprivileged children in Dhaka, including food distribution and Eid outreach.

Judge, National Business Case Study Competitions

- Judge, six case competitions at KUET: ReinforCE 1.0 “Crack the Case”, Round 2 (Concrete Society of KUET); BitFest 2025 IT Business Case Competition (Department of CSE); CASESPECS 1.0, first round (Spectrum); BizBash 2.0, 3.0 and 4.0 (KUET Career Club).

AWARDS & ACHIEVEMENTS

- Lifetime Achievement Award, Viqarunnisa Noon Debating Club, Monalisa-VDC 10th National Debate Tournament (2025)
- Hult Prize Regional Finalist: Ho Chi Minh City (2020) and Kuala Lumpur (2019); On-Campus Runner-Up (2019, 2020)
- Gold Award, UniV International Poster Competition on Achieving the SDGs (2021)
- Champion or runner-up in 20+ national poster presentation, business case, innovation and debate competitions
- HSC Board Scholarship, Talent Pool (2015)

LANGUAGES

Bangla (native) • English (fluent) • Hindi (fluent) • Korean (basic)